The Conception of a Practical Legal & Compliance Tool Through Mathematical Modeling and Statistical Analysis : Global Environmental Impact & Efficiency Valuation Model

By Habbine Estelle KIM*

ABSTRACT

Environmental awareness has significantly influenced consumer decision-making processes. Although an extensive legal, and regulatory frameworks have been developed at national and international levels, compliance remains an important topic to be addressed. The present paper deals with corporate self-regulation, transparent reporting and investor due diligence relating to activities, products, and services with an environmental impact. It outlines major enforcement challenges in practice, partly caused by the absence of a "common standard" or a "reference point" (Section I). Then, it suggests the instrumentalisation of an original mathematical model as an interdisciplinary solution (Section II). The quantitative and qualitative variables, as well as the parameters within which they are interpreted, are determined through a cost-analysis approach based on metrology considerations. The integration of comparative features within the algorithmic expression endows the tool with a practical value. The factors taken into account are based on legal and compliance, as well as economic and policy considerations. The results are translated into a numerical index which allow to position each utility case based on their weight of environmental impact, and efficiency. Subsequently, statistical tests may be made from the global dataset. This tool may serve to facilitate the identification of any accounting fraud, and the making of cost-risk assessments in the long run. It may also help examine the existence of any associations between the identified variables, reflective of the circular economy. Finally, it may enable the interpretation data for any prospective decision-making purposes from interested parties, including corporations, operators, investors, and consumers. Intergenerational and international solidarity, transparency, precaution, subsidiarity, and no harm constitute the underlying principles.

*Habbine Estelle KIM (habbine.kim@archelegal.com), dual-qualified lawyer (Paris & New York Bars). Partner (Archè SELARL), 'Legal Expert' (Station F - HEC Incubator), Regional Governor – France (International Association of Korean Lawyers), Consultative Committee Member (European Law Institute). LL.B (University College London), Masters in Law, Economics and Management - EU Law major (Paris 1 Panthéon-Sorbonne).

Table of Contents

INTRODUCTION.....3

I. LEGAL & PRACTICAL CHALLENGES TO SUSTAINABILITY PROJECTS 3

A. Legal Challenges.....3

B. Practical Challenges.....6

II. **TOWARDS A GLOBAL ENVIRONMENTAL IMPACT & EFFICIENCY VALUATION MODEL.....9**

A. Applying the First Approach – Expressing Real Environment-related Problematics and Solutions into Mathematical Terms 10

(i) Defining the Variables 11

(ii) Determining the Calculation Method 15

B. Applying the Second Approach – Deriving Practical Results Through Numerical and Statistical Analysis.....17

(i) Two-tailed Paired Samples T-test for Dependent Variables : Expected vs. Effective Data 19

(ii) Regression Analysis.....22

(iii) Chi-square Test : Association and Homogeneity.....24

(iv) One-way ANOVA N-Sample Independent Test : Utility Cases & Efficiency.....25

(v) Newcomb Benford's Law : Transparency and Prevention of Accounting Data Fraud 28

CONCLUSION 29

BIBLIOGRAPHY.....30

INTRODUCTION

Sustainability and climate action constitute important global objectives that require immediate attention.[1] Reducing the externalities, and the costs of climate change, as well as preventing the exacerbation of social injustice remain primordial.[2] The present paper seeks to offer a practical solution to foster legal compliance, and enforcement in this respect, whilst not losing grasp of the underlying economic context, and practical realities.

Section I outlines various challenges in both defining, delimiting, and determining the quantum of an environmental impact. It aims to address the legal and practical difficulties related to activities, products or services with an environmental impact in case of non-compliance. They include problematics on self-regulation, transparent reporting, investor due diligence, and proportionate sanctions.

Section II suggests an interdisciplinary approach to institute a coherent, common reference point that is understandable by all (operators, consumers, investors, policymakers, courts and lawyers). A multi-phased mathematical model is conceptualised to evaluate the global sustainability impact, and the efficiency of projects with sustainable goals. Then, certain statistical tests are made from a data analytics perspective, in order to endow both raw, and processed data in this project with a practical value.

I. LEGAL & PRACTICAL CHALLENGES TO SUSTAINABILITY PROJECTS

The present section outlines the legal (Subsection A) and practical challenges (Subsection B) in qualifying, and quantifying the environmental impact of sustainability projects. It namely addresses the difficulties relating to the fractionalisation of laws, the lack of a common reference standard, as well as access to justice and fair competition problematics in the context of enforcement, and legal compliance.

A. Legal Challenges

This subsection examines some sources of practical difficulties faced by operators carrying out pro-environmental projects.

Firstly, one of the elementary challenges to pro-environmental projects is that there is a lack of a clear and precise definition of "sustainability". The latter is an umbrella term that

[1] Article 3(3) of the Treaty on European Union (TEU); Articles 11 and Article 191 of the Treaty on the Functioning of the European Union (TFEU); Article 37 of the European Charter of Fundamental Rights; Paris Agreement; United Nations Sustainable Development Goals (SDG); World Health Organization (WHO), "*WHO issues urgent call for global climate action to create resilient and sustainable health systems*" 24.05.2023, https://www.who.int/news/item/24-05-2023-wha76-strategic-roundtable-on-health-and-climate; The Economist, "*Why tackling global warming is a challenge without precedent*", 23.04.2020, https://www.economist.com/schools-brief/2020/04/23/why-tackling-global-warming-is-a-challenge-without-precedent ; Mukherji, A., et al. "*Synthesis Report Of The IPCC Sixth Assessment Report (AR6).*", 2023, United Nations;

[2] Lenton, Timothy M., et al. "*Quantifying the human cost of global warming*." Nature Sustainability (2023): 1-11.

does not merely encompass environmental factors,[3] but also includes socioeconomic considerations, amongst others. In the absence of set conditions, standards or a common definition, it is difficult to identify a comprehensive ecosystem, index, system of governance, central authority, common reference point, conditions or requirements that are capable of classifying, qualifying, recognising, certifying or validating projects, businesses, structures, products or services as being sustainable.

At the European Union ("EU") level, the following terms may help englobe or narrow down its scope : 'green', 'environmental', 'circular economy' or 'eco-friendly'.[4] However, there exists a gap between their commercial, technical, legal, regulatory, and lay meanings. Certain practical use of the terms are susceptible to generate confusions. For example, not all of these key terms imply the existence of marks, labels, packaging or certifications, which may be mandatory or voluntary depending on whether the relevant goods or services are subject to any laws, and regulations.[5] This is why at the EU banned greenwashing practices, such as the making of generic environmental claims, and the apposition of sustainability labels without approved certification schemes or established by published authorities.[6]

---

[3] Watling, A. et al. "*Attitudes towards Sustainability: A Quantitative Study of Sustainable Ålidhem*". 2011, Umea Universitet, Umea, Sweden. http://umu.diva-portal.org/smash/record.jsf?pid=diva2%3A430152&dswid=-6754 (On how sustainability essentially refers to the recurrent factors of economic viability, social justice and environmental impacts that contribute to a durable solution); United States Environmental Protection Agency, "*Learn About Sustainability*", https://www.epa.gov/sustainability/learn-about-sustainability#what ("*Sustainability is based on a simple principle: Everything that we need for our survival and well-being depends, either directly or indirectly, on our natural environment. To pursue sustainability is to create and maintain the conditions under which humans and nature can exist in productive harmony to support present and future generations*."); Tucker, D.T., Stanford University, "*What is Sustainability? A definition by environmental scientist Pamela Matson*", 05.21.2018, https://earth.stanford.edu/news/what-sustainability-definition-environmental-scientist-pamela-matson ("*Sustainability is about the ability to meet the needs of people and their communities and organizations not just in the near term but over the long term. It's about people and their needs, not just about technology, the environment or "being green." With an environmental lens, sustainability is about managing and protecting Earth's natural resources, ecosystems, climate and atmosphere so that current generations and future generations will have the things they need to live a decent life. In doing so, the millions of other species with whom we share the planet will also benefit*."); Moore, J.E., et al. "*Developing a comprehensive definition of sustainability*", 2017, Implementation Science, 12(1), pp.1-8.

[4] European Parliamentary Research Service (EPRS), G. RAGONNAUD, Briefing, EU Legislation in Progress, "*'Green claims' directive Protecting consumers from greenwashing*", 05.10.2023, https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2023)753958; Council of the European Union, "*Proposal for a DIRECTIVE OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL amending Directives 2005/29/EC and 2011/83/EU as regards empowering consumers for the green transition through better protection against unfair practices and better information*", 2022/0092(COD); European Parliamentary Research Service (EPRS), N. SAJN, "*Empowering consumers for the green transition*", 16.06.2022, https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2022)733543

[5] International Trade Administration, "*EU Country Commercial Guide*", 11.08.2022, https://www.trade.gov/country-commercial-guides/eu-labelingmarking-requirements ("*A mark is a symbol and/or pictogram that appears on a product or its respective packaging. These range in scope from signs of danger to indications of methods of proper recycling and disposal. The intention of such marks is to provide market surveillance authorities, importers, distributors, and end users with information concerning safety, health, energy efficiency and environmental issues relating to a product. Labels, on the other hand, appear in the form of written text or numerical statements, which may be required but are not necessarily universally recognizable. Labels typically indicate more specific information about a product, such as measurements or an indication of materials that may be found in the product (such as in textiles or batteries*).)

[6] European Parliament, Press Release, "*EU to ban greenwashing and improve consumer information on product durability*", 19.09.2023, https://www.europarl.europa.eu/news/en/press-room/20230918IPR05412/eu-to-ban-

Taking the aforementioned factors into account, it may be argued that the notion of sustainability at large requires the adoption a flexible, holistic approach. However, this does not necessarily mean that a one-fit-for-all approach is practically viable, nor that it is actually possible to devise one master solution covering every industry, sector, goal, and operator. Due to the need to include emerging markets and fast-evolving technological progressions, economic feasibility, market maturity, as well as interdependencies existing between certain industries, the fractionalisation of laws and regulations, and the multiplicity of technical standards, are inevitable.

In order to reduce the abovementioned discrepancies, many practical, legislative and regulatory endeavors have already been made at national, European and international level.[7] Attention has been drawn to the reliability, repeatability and reproducibility of certain tests and impact assessments, the framework and methods in which they may be conducted, as well as the precision, and accuracy of the results that may be obtained.[8] The EU strived to promote further clarity, rationality, and simplification in favour of a more harmonized system of reference.[9] Nevertheless, what is essentially lacking in practice is a bridge of communication setting a common referential ground of understanding for the Member States, Third Countries, as well as with the national authorities, operators, investors, and consumers. So long as this is left resolved, hardships will persist in relation to preserving and fostering innovation, competitivity, consumer choice, and consumer trust.

---

greenwashing-and-improve-consumer-information-on-product-durability

[7] European Commission, "*Corporate sustainability reporting*", https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en; Directive (EU) 2022/2464; Climate Benchmarks Regulation EU 2019/2089 and 2020/1818; Sustainable Finance Disclosure Regulation EU 2019/2088; Taxonomy Regulation EU 2020/852; Measuring Instruments Directive 2014/32/EU; Directive on Units of Measurement 80/181/EEC ; Copernicus Climate Change Service, ECMWF, European Commission, "*Essential climate variables for assessment of climate variability from 1979 to present*", published on 07.11.2018, updated on 05.10.2023, https://cds.climate.copernicus.eu/cdsapp#!/dataset/ecv-for-climate-change?tab=overview; Balvanera, P., et al. "Essential ecosystem service variables for monitoring progress towards sustainability", 2022, Curr. Opin. Environ. Sustain., 54, 101152. ; United Nations Climate Change, "Overview; Implementation of systematic observation", https://unfccc.int/es/node/15979; Joint CEOS/CGMS Working Group Climate, Schulz, J. et al., "Statement of Space Agency Contributions in Support of the Paris Agreement", 16.11.2018,

[8] Brown, M.T., et al., "*Predicting national sustainability: The convergence of energetic, economic and environmental realities*", 2009, Ecological Modelling, ELSEVIER, 220(23), pp.3424-3438.; Munda, G. "*Measuring Sustainability: A Multi-Criterion Framework. Environment Development and Sustainability*." 2005 7. 117-134. 10.1007/s10668-003-4713-0.; Ibáñez-Forés, V. et al. "*Sustainability reports as a tool for measuring and monitoring the transition towards the circular economy of organisations: Proposal of indicators and metrics*", 2022, Journal of Environmental Management, Volume 320, 115784, ISSN 0301-4797, https://doi.org/10.1016/j.jenvman.2022.115784,https://www.sciencedirect.com/science/article/pii/S0301479722013573; UCL (Department of Geography) and Center for Global Environmental Research, National Institute for Environmental Studies CARTER, T.R., et al. "*IPCC Technical guidelines for assessing climate change impacts and adaptations*." 1994, Intergovernmental Panel on Climate Change, Part of the IPCC Special Report to the First Session of the Conference of the Parties to the UN Framework Convention on Climate Change, World Meteorological Organisation/United Nations Environment Programme, CGER-I015-94

[9] European Council, Council of the European Union, "*Timeline - European Green Deal and Fit for 55",* https://www.consilium.europa.eu/en/policies/green-deal/timeline-european-green-deal-and-fit-for-55/; European Parliament, Policy Department for Citizens' Rights and Constitutional Affairs Directorate-General for Internal Policies, Prof. Dr. Lorenzo SQUINTANI "Study on the Harmonisation of EU Environmental Law" PE 705.033 – February 2022

A second major hurdle comes with the plurality of amendments, extensions or the advent of new legal and regulatory dispositions. It becomes increasingly difficult for certain companies (especially SMEs with limited resources and individual contributors) to consistently track and ensure legal, and regulatory compliance.

B. Practical Challenges

This subsection exposes several practical impediments to pro-environmental projects. One of the difficulties results from the plurality of sources of compliance.[10] Indeed, there exist additional hardships raised by the diversity of standards, taxonomies and calculation methodologies (legal, technical, industrial, commercial). This is more so, where the applicable standard, and point of reference may differ based on whether the scope of compliance is set at national, European or international level. For example, there exist European Regulations[11] 2018/858/EU[12], 2019/2144/EU[13], 117 UNECE, 2020/740/EU and national dispositions forming the legal basis for eventual investigation, control or litigation[14] for products and services involving energy performance, with potential health and safety implications. In parallel, there are International Organisation for Standardisation (ISO) standards in relation to environmental management standards, health and safety standards, and energy management standards.[15] There may be additional industrial, sector-specific, or technical standards beyond those established by ISO.[16] Although standards remain voluntary, unless rendered mandatory by adopting legislation, certain operators take an extra step to seek certification by having accredited third parties audit compliance to. Added to these are guidelines and reports, staff working documents, studies, communications and calculators, amongst other documents available at the European Commission's website.

---

[10] Grimm, J.H., Hofstetter, J.S. et al. "*Exploring sub-suppliers' compliance with corporate sustainability standards*", 2016, Journal of Cleaner Production, 112, pp.1971-1984; Vigneau, L., et al. "*How do firms comply with international sustainability standards? Processes and consequences of adopting the global reporting initiative*", 2015, Journal of business ethics, 131, pp.469-486.

[11] They envisage various testing methods and considerations for calculating the tire fuel efficiency and other parameters (rolling resistance, wet grip, rolling sound emissions) and related labeling based on a conversion of quantitative technical results into a designated Class

[12] Regulations 2018/858/EU, 2019/2144/EU

[13]Regulations 2018/858/EU, 78/2009/EC, 79/2009/EC, 661/2009/EC, 631/2009/EC, 406/2010/EU, 672/2010/EU, 1003/2010/EU, 1005/2010/EU, 1008/2010/EU, 1009/2010/EU, 19/2011/EU, 109/2011/EU, 458/2011/EU, 65/2012/EU, 130/2012/EU, 347/2012/EU, 351/2012/EU, 1230/2012/EU and 2015/166/EU

[14] In relation to prohibited forms of commercial practices, the French Consumer Code (Articles L121-1 to L121-4, L132-1 to L132-2, L531-6, R531-3 and L521-7) contains dispositions related to any allegations, indications, information or presentations, essential or principal characteristics, substantial qualities, properties and results in relation to a good or service, as well as French law n°2008-3 transposing Directive 2005/29/EC, later modified by Directive 2019/2161/EU

[15] ISO 32210; ISO 14007 (Guidelines for determining environmental costs and benefits); ISO 14008 (Monetary valuation of environmental impacts and related aspects Applied in a Case Study for a Stock Holding Investment Company); ISO 14097 (Standard for Investment, Financing and Climate Change)

[16] European Sustainability Reporting Standards (ESRS); Corporate Sustainability Reporting Directive (EU) 2022/2464 ; Non-Financial Reporting Directive (NFRD) 2014/95/EU ; International Organization of Securities Commissions, "IOSCO endorses the ISSB's Sustainability-related Financial Disclosures Standards" IOSCO/MR/19/2023, 25.07.2023

It is worth noting that projects of significant coverage, technicity and complexity, namely those involving energy and transport, are operated at national or international scale with the involvement of public entities. For example, in procurement projects involving a consortium often imply the existence of (i) an element of extraneity (due to applicable law or jurisdiction that may go beyond the national territory), and (ii) multiple operator (including agencies, subcontractors, intermediaries, distributors and retailers).

Compliance obligations are often agreed horizontally between partnering entities and then handed downstream to their subcontractors, at national or international scale, starting from the "head operator" of the overall organization structure. The applicable law, jurisdiction and standard are likely to be affected by the bargaining power, as well as the economic, political or other forms of power that the head operator may have over others, as well as national/international public policy considerations, where deemed relevant. If operators in different countries are in charge of contributing a specific part (e.g. a spare part) for a wider product, as is the case for the ITER project, the question remains as to which standard would apply to the global product, and how to deal with cases of conflicting or multiple standards.

Further, projects involving public and/or private investment (e.g. startups, innovative projects, etc.) necessarily involve transparency and declarative obligations subject to deadlines, and penalties. Here, there may be difficulties relating to time-keeping (e.g. requesting and collecting data from all the subcontractors that may have different calculation/assessment standards), confidentiality concerns, as well as other practical business concerns (e.g. language barriers, industry practice of not having "paper" contracts but verbal agreements formalised by execution based on mutual trust).

In any case, where divergent points of reference exist, the final result or conclusion obtained derive from an approved equivalent, an average, or a conversion. This risks heightening bias and error rates, reducing accuracy, and creating inconsistent result interpretations. This is especially so if the results are precise but not accurate due to a restrictive data range, or if the conversion is made from a quantitative to a qualitative result.

Experts and professionals ought to be consulted to determine which laws, rules and regulations apply to the relevant activities, and goods. Subsequently, the actual compliance of existing practices ought to be verified. All necessary measures would have to be put in place (including prevention, security, remedial, mitigation, and other compliance measures) in this respect. The costs of compliance would be borne by the company. The more innovative, interdisciplinary or regulated the good or service in question, the more complex, time-consuming, and costly the compliance assessment would likely be. This complexity is heightened in international settings where multiple or conflicting standards may apply across various Member States or Third Countries.

The aforementioned uncertainties are likely to affect companies that are exposed to investigations, controls, sanctions or litigation. Such risks are likely to have a critical impact on the companies' operations, decision-making, strategies, and survival. They also call for added costs to the liabilities of companies (additional tests and assessments before accredited laboratories, experts or other professionals, legal fees, enhanced prevention, remedial or mitigation measures, corrective measures, penalties, sanctions or other consequences), some of

which may be critical or have a significant impact on the business.

Since not all operators function in perfect conditions, in consideration of available resources of all types (human, capital, natural, physical, IP, financial, time), the focus ought to be ideally put at the initial phase of project implementation. However, in practice, not all companies and businesses can afford to wait for the theory to be perfect to initiate any action, namely to acquire competitive advantage, preserve competitivity, innovate, and generate profits. If too much focus is put at the investigation, control, remedies and sanction stages, there may be significant financial and commercial impact to companies. This would especially be the case if the alleged non-compliance is minor or does not have a significant impact, or causes negligible real prejudice.

Finally, although there are many tools of measurement, it remains difficult in certain instances to compute and delineate the scope and the degree of environmental harm, whether qualitatively or quantitatively, if measurable at all *ab initio*.[17] This may be due to the lack of scientific, technological and/or technical knowledge, lack of transparency and sufficient data, or the multiplicity of measuring standards, methodologies and techniques worldwide, as well as the heterogeneous nature of the test samples, amongst others.

In light of the above concerns, a mathematical model is created to enable the comparing and contrasting of predicted and actual performance of the operators, as well as their efficiency (Section II).

[17] Das, A., et al., "*How do companies measure and forecast environmental impacts when experimenting with circular business models?*", Sustainable Production and Consumption, 2022, Volume 29, pp.273-285.; Khan, S.A.R., et al., "*Measuring the impact of renewable energy, public health expenditure, logistics, and environmental performance on sustainable economic growth*", Sustainable development, 2020, Volume 28(4), pp.833-843.

II. Towards a Global Sustainability Impact & Efficiency Valuation Model

The present section aims to introduce a Global Sustainability Impact & Efficiency Valuation Model ("Model") that integrates a dynamic, cost-based algorithm into a centralised system. In this respect, it is reminded that sustainability engages a process of creating and durably maintaining conditions that allow current and future well-being of a community, nation, or planet. For the purpose of this section, a particular interest will be given to the sustainable action and performance of businesses. Considerations will be given to environmental science and sustainability economics, namely the notions of discounting, abatement, irreversibility, and the global commons.[18]

Prior to entering into the details of the mathematical modeling and interpretation, it is worth noting that the role and purpose of applied mathematics and mathematical modeling are subject to certain identifiers. The latter vary according to the initial approach adopted to construe the Model.

Some contend that mathematical modeling consists in the numerical translation and expression of real life problematics, let a "*conversion activity of a real problem in a mathematical form. Modeling involves to formulate the real-life situations or to convert the problems in mathematical explanations to a real or believable situation*"[19] ("First Approach"). This approach would require that a target practical problematic be first identified to later determine the variables, the rules, as well as the applicable model, and reasoning. It aims to best reflect the reality, and illustrate by other means a system that exists.

Others adopt a more purposive approach of numerical analysis, and suggest that mathematical models can help "*[develop scientific understanding through quantitative expression of current knowledge of a system ; test the effect of changes in a system; and aid tactical and strategic decision making]*"[20] ("Second Approach"). Depending on the initial understanding, reason and conception behind the use of a mathematical model, the derived result (and thus the conclusion subsequent to the numerical analysis) would differ. This approach would require that the end goal or the object of interest be first identified to determine any such elements. It aims to find solutions and seek further understanding, and insight into unresolved problematics.

---

[18] Guest, R. "*The economics of sustainability in the context of climate change: An overview*", 2010, Journal of World Business, Volume 45, Issue 4, 2010, ISSN 1090-9516, https://doi.org/10.1016/j.jwb.2009.08.008.

[19] Dündar, S et al., "*Mathematical Modelling at a Glance: A Theoretical Study. Procedia - Social and Behavioral Sciences*", 2012, 46. 10.1016/j.sbspro.2012.06.086; Jankalová, M., et al. "*Sustainability Assessment Using Economic Value Added*", 2020, *Sustainability* 12, no. 1: 318. https://doi.org/10.3390/su12010318 (Economic Value Added methodology is based on financial and non-financial indicators considering the financial situation of companies and to calculate the actual economic profit of a company)

[20] Glenn M. et al., "*An Introduction to Mathematical Modelling*", 2008 ; Fowler, A.C., "*Mathematical models in the applied sciences*", 1997, Volume 17, Cambridge University Press.; Porter, M.E. et al. "*Creating Shared Value*". Harv. Bus. Rev., 2011, 89, 62–77F (Figge and Hahn published their first Sustainable Value Added (SVA) model based on comparing earned value of a company with a benchmark entity, assuming the same impact of both the company and the benchmark entity on the environment)

Overall, there exist various approach and reasons to mathematically test and model climate science, and impact. [21] However, not all take into account the considerations of public policy, laws and regulations, as well as other socioeconomic factors likely to influence the results. Many are industry, sector or subject-specific at micro level, restricted to a pre-determined set of data. Furthermore, many remain either theoretical or practical, but not both. As such, the solution proposed in the present paper is contextualised concretely in the legal compliance framework but practically translated in a holistic, flexible manner via a mathematical model which will combine both approaches outlined above.

The first part outlines the specifics on the construction of the Model (A). The second part identifies the use and value of the Model (B).

A. Applying the First Approach – Expressing Real Environment-related Problematics and Solutions into Mathematical Terms

This sub-section outlines the methodology in structuring the Model in application of the First Approach. First, it is worth noting that comprehensive and flexible model is constructed by taking into account the circular and sustainable ecosystem in which one or more environment-related projects are deployed, as well as metrological perspectives[22].

On the basis of this structural conception of the Model, the framework and the scope of application of the Model may be determined. The legal and compliance scope will be limited to those that contribute to the 17 United Nations sustainable development goals as outlined in its 2030 Agenda, albeit classified horizontally (per industry and sector), and vertically (per action point categories and individualised solutions), with a particular focus on the environmental and economic aspects of sustainability (e.g. sustainable cities and communities, responsible consumption and production, climate action, life below water, life on land, clean water and sanitation, affordable and clean energy, and industry, innovation, and infrastructure).

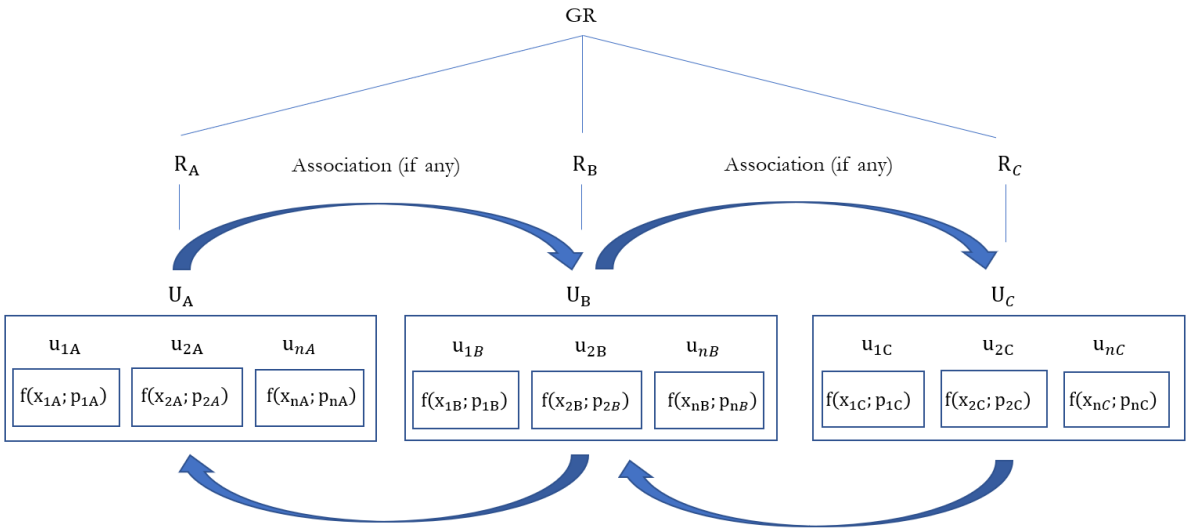
Mathematically, the algorithmic expression is determined by considering the structural aspect of existing ecosystems, going top-down from a generic category of issues to more specific ones. In brief, the first level provides for one result obtained for one individual, defined project, for a specific issue. The second level aggregates the individual results obtained at the

---

[21] Dündar, S et al., "*Mathematical Modelling at a Glance: A Theoretical Study. Procedia - Social and Behavioral Sciences"*, 2012, 46. 10.1016/j.sbspro.2012.06.086

[22] Plant, A.L. et al. "*Reproducibility and replicability in science, a metrology perspective*", 15.06.2018, National Academies of Sciences, Engineering and Medicine Committee on Reproducibility and Replicability in Science Report. (Reproductibility means "*Precision in measurements under conditions that may involve different locations, operators, measuring system s, and replicate measurements on the same or similar objects. The different measuring systems may use differen t measurement procedures*."; Repeatability means "*Precision in measurements under conditions that include the same measurement procedure, same operators, sa me measuring system, same operating conditions and same location, and replicate measurements on the same or similar objects over a short period of time*.")

first level, and classifies them based on a common category of issues to which the diverse, individual solutions are proposed. The third level aggregates the results obtained at the second level, without any categorical segregation, and allows to visualize the overall impact of the projects deployed globally in relation to the environment. Therefore, are identified at each level a solution to a problem. These elements may be arranged, and combined so as to best reflect the existing ecosystem. The diagram hereinbelow provides a visual overview of the overall Model, integrating the principle of circular economy. The dimension covered is temporal, spatial, societal, and economic.



The present Model takes into account the imperfectness and bias that may exist between the multiple data sets. The large scope of the global result would have cut down outliers, inaccuracies, and unreliable data. In order to reduce the variability and randomness of the dataset further, the calculation method is based on a range for each variable (per relevant unit, scale and sensitivity desired). Structural bias and uncertainties are controlled by limiting the scope of the input and output to pre-determined cost-factors, applicable to individual sets (the combination of all the individual use cases classified inside each category divided in terms of common objective) as well as the universe (all the use cases combined in all the categories).

(i) Defining the Variables

Generally, the determination of the component variables takes into account the aim of neutralising and standardising the complex technical reference points that lead to result conversions or interpretations based on the divergent national, European, and international technical standards.

In essence, there will be three sets of data serving as the input value: minimal predicted data, maximum predicted data and real data. The type of data collected is widely construed in order to preserve sufficient flexibility for its future application. Nevertheless, they remain sufficiently identifiable, definable and may be delineated to ensure understanding of the Model.

In order to maintain coherence of the global calculation rationale, all the variables are expressed, and evaluated in terms of their predictive costs in euros. Each category of relevant data is denominated as "cost factors". Various cost factors may be taken into account in the predictive cost assessment. They include commercial, legal and technical costs, including, but not limited to the total cost of research and development, conception, design, implementation, adaptation, repair, maintenance, replacement, rectification, destruction, storage, human resources, facilities, space, the capacity of the operator (available funds, facilities and resources including internal, external, private and public funds at all levels), and the capital use (fixed or total assets of a company), expenses and expenditures, disbursements, mitigating factors (efficient remedy, rectification), aggravating factors (recurrence, gravity, large scope of impact and durability of impact of the harm) for a determined term, in relation to an individual utility case. In case of conversion of currency, the applicable exchange rate is that on the date of complete registration of data. The final result is expressed in terms of a quantitative index.

Other associated factors may be taken into account depending on the relevant sector, industry and entity implementing the individual utility case, such as the territorial scope (national, EU, international), the reasonable constant range (degree of sensitivity of the results), and the time interval or target time objective. More specifically, the variables of the Model are set and defined in the following manner.

Let there be qualitative indicators (quantitatively expressed in terms of costs), and quantitative indicators (the efficiency coefficient expressed in terms of a value range between 0 and 1, which may be modulated to heighten sensitivity),

Such that:

u_n = individual utility case, a qualitative indicator representing one solution in relation to a common category of solutions constituting one set (U_n).

For example, if the set U_A relates to the common objective of tackling deforestation, it may be constituted by an assembly of the following individual solutions:

u_{1A} = planting trees
u_{2A} = preserving natural resources
u_{3A} = creating infrastructures against natural disasters

This qualitative indicator will quantitatively be translated in terms of its cost of implementation (x) and the efficient coefficient (p) :

x_n = individual input variable (expressed in terms of cost in euros)

For example,

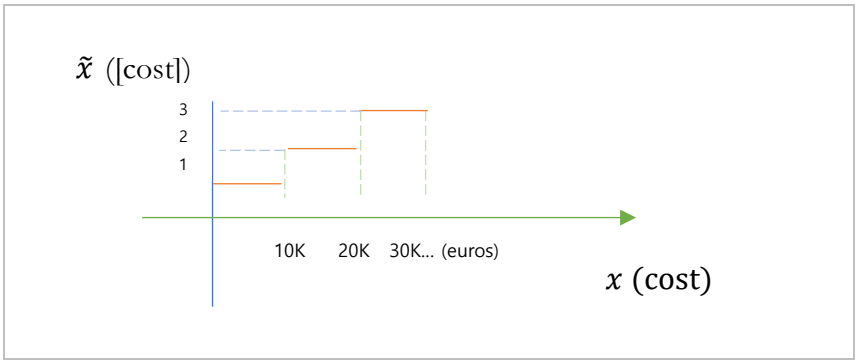
x_{1A} for u_{1A} (tackling deforestation)	= 1 500 €
x_{2A} for u_{2A} (preserving natural resources)	= 28 000 €
x_{3A} for u_{3A} (infrastructures against natural disasters)	= 1M €

Cases of total inaction ($x_n = 0$) shall be omitted in this valuation model.
These input costs can vary in a wide range, from hundreds to millions of euros, which makes them hard to interpret. We may instead aggregate them in brackets of, say, 10 000 euros. The cost bracket $\tilde{x}$ is obtained from the cost as :

$\tilde{x}$ (cost bracket)	x (cost in euros)
1	$0 \leq x < 10\,000$
2	$10\,000 \leq x < 20\,000$
3	$20\,000 \leq x < 30\,000$
…	…

Costs exceeding a certain level x_{max} are allocated to the last bracket. Defined this way, the cost bracket is computed using the following formula:

$$\tilde{x} = \min\left(\left\lfloor \frac{x}{10\,000} \right\rfloor + 1, \left\lfloor \frac{x_{max}}{10\,000} \right\rfloor + 1\right)$$



r_n = individual result, a quantitative indicator obtained for a given individual utility case (u_n).

For example,

r_{1A} for u_{1A} (planting trees)	= 0.2
r_{2A} for u_{2A} (preserving natural resources)	= 0.4
r_{3A} for u_{3A} (creating infrastructures against natural disasters)	= 0.5

U_n = global utility case, a qualitative indicator equal to the assembly of all the individual utility cases (u_n) for a common category of solutions, constituting one set.

For example,

U_A = tackling deforestation
U_B = renewable energy
U_C = waste management

R_n = category result, a quantitative indicator obtained for a given individual utility case (U_n).

For example,

R_{1A} for u_{1A} (tackling deforestation) = 0.05
R_{2A} for u_{2A} (renewable energy) = 0.5
R_{3A} for u_{3A} (waste management) = 0.2

GR = global result for the universe, a quantitative indicator for the aggregation of all the global utility cases (U_n), expressed quantitatively as an aggregation of all the individual results (R_n).

Then, the Utility Performance Efficiency Index is defined as follows.

Let p_n = utility performance efficiency coefficient per individual utility case (u_n).

P takes into account the total cost of the negative externality as well as the duration of the project compared to European objectives.

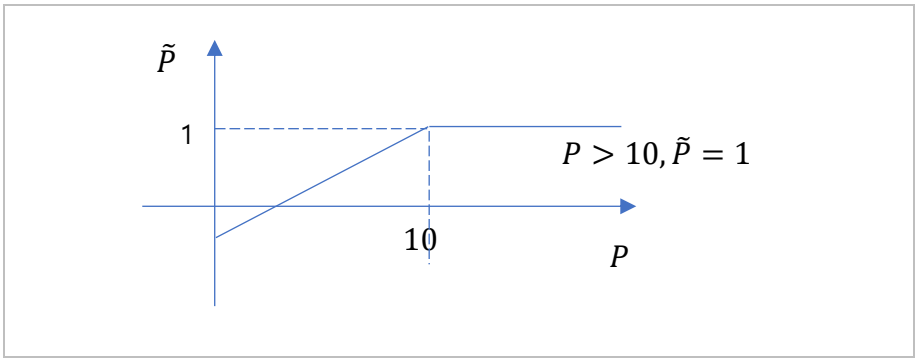
For example,

It may take Germany 27 years from 2023 to 2050 and cost up to 900 billion euros to put in place the envisaged climate action.

And $\tilde{p} = \min\left(\frac{p}{10}, 1\right)$

$\lvert\tilde{p}\rvert$ (efficiency bracket)	p (efficiency coefficient)
1	$0 \leq p < 0.1$
2	$0.1 \leq p < 0.2$
3	$0.2 \leq p < 0.3$
…	…
10	$0.9 \leq p < 1$

The closer the value is to 0, the more inefficient or negligible the efficiency impact on R.



For purposes of readability or ease of interpretation, p may be set as natural numbers, going from 0 to a maximum value P, rather than using decimals.

Otherwise, the sensibility of p could be increased by amending the range or the frequency of p. In this respect, three decimal digits may be used instead of one decimal digit: $\tilde{p} = \min\left(\frac{p}{1000}, 1\right)$.

(ii) Determining the Calculation Method

For a given individual utility case (u_n), we combine the utility performance efficiency coefficient (p_n), and the input cost (x_n) into an individual indicator (r_n) :

$$r_n = f(x_n, p_n) = e^{-x_n p_n}$$

Defined this way, the individual indicator (r_n) decreases whenever cost or efficiency increase.

What is more, the individual indicator (r_n) for each individual utility case (u_n) of a given set A can be combined multiplicatively, so that their effects compound to a global category result (R_A) which is expected to reach a point closer and faster to the optimal point 0 than for each r_n taken individually :

$$
\begin{aligned}
R_A \quad &= \quad - \left| f(x_1; p_1) \, x \ldots f(x_2; p_2) \ldots x \, f(x_n; p_n) \right| \\
&= \quad - \left| e^{-x_1 p_1} \, x \, e^{-x_2 p_2} \, x \ldots x \, e^{-x_n p_n} \right| \\
&= \quad - e^{-(x_1 p_1 + x_2 p_2 + \cdots + x_n p_n)} \\
&= \quad - e^{-\sum_{k=1}^{n} x_k p_k} \quad \xrightarrow[n \to +\infty]{} \quad 0
\end{aligned}
$$

This aggregate category result for set A (R_A) depends on the total budget of (efficiency-adjusted) input costs in that set $\sum_{k=1}^{n} x_k p_k$.

The compounded effect of every global category results (R_n) would yield a global result (GR):

$$
\begin{aligned}
GR \quad &= \quad - \left| f(x_1; p_1) \, x \ldots f(x_2; p_2) \ldots x \, f(x_n; p_n) \right| \\
&= \quad - \left| e^{-x_1 p_1} \, x \, e^{-x_2 p_2} \, x \ldots x \, e^{-x_n p_n} \right| \\
&= \quad - e^{-(x_1 p_1 + x_2 p_2 + \cdots + x_n p_n)} \\
&= \quad - e^{-\sum_{k=1}^{n} x_k p_k} \quad \xrightarrow[n \to +\infty]{} \quad 0
\end{aligned}
$$

For which the global result (GR) would be <u>inferior to</u> :

(i) the value of each individual indicator (r_n) for each individual utility case (u_n); <u>and</u>

(ii) the value of each global category indicator (R_n) for each global utility case (U_n)

Such that :

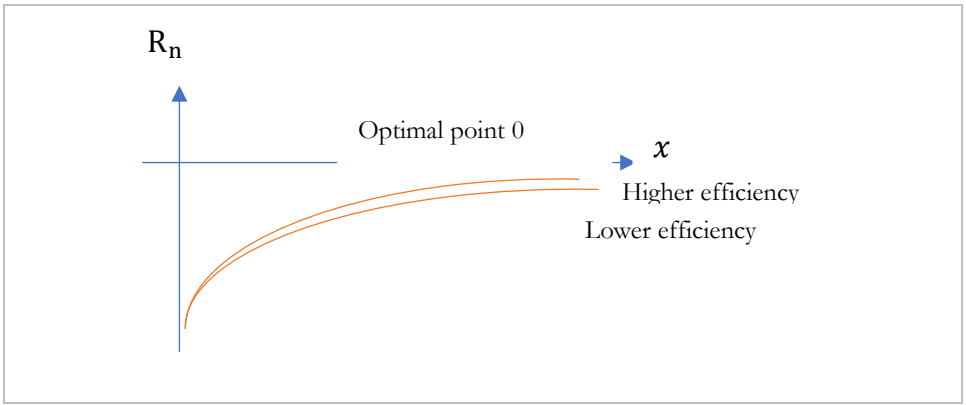
$$
\begin{aligned}
R_n \quad &< \left| f(x_1; p_1) \right| \\
R_n \quad &< \left| f(x_2; p_2) \right| \\
&\ldots \\
R_n \quad &< \left| f(x_n; p_n) \right|
\end{aligned}
$$

Since the resulting graph shows a harm test, the optimal point for the global result (GR) shall be 0, representing level 0 of harm. 0 is merely a theoretical optimal point, since there exist negative externalities that cannot be undone. As such, the purpose would be to approach closest to 0 at the earliest point in time such that:

(i) The aggregation of each individual utility case result (r_n) must yield a category utility case result (R_n) that is closer to 0 than each individual utility case result taken separately.

(ii) The aggregation of each category utility case result (R_n) must yield a global result (GR) that is closer to 0 than each category utility case result taken separately.

The premise is that the category result (R_n) and global result (GR) must tend towards infinity. As such, the sum of all the results associated to the utility cases, taken individually of combined, must not be so negligible so as to give a finite result. In other words, each individual utility case (u_n) and global utility case (U_n) does not have a negligible weight of impact.



B. Applying the Second Approach – Deriving Practical Results Through Numerical and Statistical Analysis

This sub-section examines the different hypothesis and conclusions that may be derived in application of the Second Approach, considering that the Model is construed to represent a certain logical process of numerical evaluation, namely relating to the following two main objectives.

The first part of the statistics and data analysis is conducted at micro-level to ascertain the feasibility, reasonableness, economic viability, efficiency and performance of one or more environment-related projects. The Model reflects a monitoring and self-assessment system based on real, quantitative figures relating to the costs engaged to create, implement, operate, amend, annul and suspend projects, as well as to make more accurate and precise predictions in the future. Due to the way in which the Model is structured, it is possible to compare the predicted costs and the real costs (financial and accounting data tracking the cash flow) engaged for a defined project. The progressive basis of comparison prevents tampering with data or having inconsistencies stemming from multiple of testing methods or artificial manipulations made by the declarer. The Model thus allows for the automation of monitoring processes in the long run. It would enable operators to examine the level and scope of impact and the achievability of a project, as well as the recurrence of certain types of costs or problems. This would help the identification of priority, urgent or major intervention areas and the assessment of the need to readjust budgets or optimise strategies so as to enhance competitivity, and preserve innovation. Consequently, it would incite self-regulation, compliance and regular, transparent reporting. Investors would have access to more reliable and structured information compiled in the long run, enabling them to assess the operator or the project's economic

viability, trustworthiness and efficiency. Naturally, this would lead to the enhancement of consumer welfare and consumer protection, namely in relation to products or services related to commercial practices with sustainability goals.

The second part of the statistics and data analysis is conducted at macro-level to derive the impact and correlation of an individual or an aggregate of environment-related projects. The collected data may serve generally to identify a general pattern. Then, the system of calculation makes it possible to identify and to provide further insight on the commonalities existing between sustainability-related problematics in different areas, industries and sectors. By reverse-engineering the results and collected datasets, it is possible to examine the existence of any association, interdependencies, confounding and causation (e.g. between individual utility cases within a global utility case context, or between individual results with regard to the global result), the efficiency per project, company, industry and sector for a common goal, or and the weight of each project, as well as proportionality (based on impact, size, efficiency, etc.). A comparative data analysis may be made for the benefit of the investors as part of their due diligence, as well as the competent national authorities in case of investigation, and control.

Having a reasonable overview over the global circumstances that surround the target project would help assess realistically, and all things considered, the most proportionate sanction applicable to the company, including its amount. For example, the "*reasonable or best endeavours*"[23] made by the company to ensure compliance or respond effectively through remedial or mitigatory[24] measures may be derived from the collected data (e.g. the costs engaged to rectify a labeling error), as well as the profits gained and the surrounding circumstances that may have affected non-compliance (e.g. lack of staff, inaction of a subcontractor, *force majeure* etc.). The benefits to the society and the environment may also be assessed based on this quantitative indicator.

---

[23] *Bulman v Fenwick & Co* [1894] 1 QB 179; Sheff*ield District Railway Co v Great Central Railway Co* [1911] 27 TLR 451; *Electricity Generation Corporation (t/as Verve Energy) v Woodside Energy Ltd* (2014) 251 CLR 540; *MUR Shipping BV v RTI Ltd* [2022] EWCA Civ 1406, [2022] WLR(D) 427, [2023] 1 All ER (Comm) 501, [2023] Bus LR 355, 205 Con LR 1; Rhodia International Holdings Ltd. v Huntsman International LLC [2007] EWHC 292 (Comm); *RAE Lambert Co UK Ltd. V Enron Europe Ltd* [1997] CLC 329 ; *CPC Group Ltd v Qatari Diar Real Estate Investment Co* [2010] EWHC 1535 (Ch) ; *Yewbelle v London Green* [2007] EWCA Civ 475 ; *UBH (Mechanical Services) Ltd v. Standard Life Assurance Co. T.L.R., 13 Nov. 1986 (Q.B.)*

[24] *Vereniging Milieudefensie v Royal Dutch Shell PLC,* the Hague District Court, C/09/571932 / HA ZA 19-379, ECLI:NL:RBDHA:2021:5337 *("The Hague District Court has ordered Royal Dutch Shell (RDS) to reduce the CO2 emissions of the Shell group by net 45% in 2030, compared to 2019 levels, through the Shell group's corporate policy."); The State of the Netherlands (Ministry of Economic Affairs and Climate Policy) v Stichting Urgenda,* The Supreme Court of the Netherlands, 19/00135 *("Climate case Urgenda. Human rights. ECHR. United Nations Framework Convention on Climate Change. Judicial order to the Dutch State to take measures against climate change. Scope of protection Art. 2 and 8 ECHR; positive obligations. Effective remedy Art. 13 ECHR. Partial responsibility states. Art. 3:305a Dutch civil code (collective action) and Art. 34 ECHR. Target of 25-40% reduction of the emission of greenhouse gases and the necessity of this target.")*

(i) Two-tailed Paired Samples T-test for Dependent Variables : Expected vs. Effective Data

An important aim of statistics is to make informed decisions based on data. Therefore, it is useful to address different statistical testing and analysis methods to interpret data. Each test will have its own correlative hypothesis (or significance) test.[25] The present sub-sub-subsection seeks to test the effectiveness of the use case by comparing expected and effective data. A paired-sample t-test may be used based on the expected ($\mathbf{u_{nAEXP}}$) and the effective individual result ($\mathbf{u_{nAEFF}}$) for each individual utility case ($\mathbf{u_n}$) corresponding to the same type of projects within the same category, i.e. the same projects carried out by different competitors (e.g. planting trees).[26] The sample data would consist of the expected and effective costs engaged for each individual utility case (u_{1A}, u_{2A}, u_{3A} … u_{nA}) (planting trees) for the same categorical utility case (U_A = tackling deforestation). The notion of "costs" may either be taken in their totality or per unit component.

[25] A "*null hypothesis*" is a statement or assumption that there is no significant difference or relationship between two variables. The purpose of the null hypothesis is to provide a baseline against which the results may be compared. Statistical tests may then serve to determine whether the results are likely to be random or whether they are statistically significant and support the alternative hypothesis. In other words, the null hypothesis is the default position of reference that is sought to be disproved. If the null hypothesis cannot be rejected, then it is deemed that there is insufficient evidence to support the alternative hypothesis. An "*alternative hypothesis*" is a statement or assumption that contradicts the null hypothesis. It represents the alternative explanation or the effect that is sought to be established. The purpose of the alternative hypothesis is to provide an alternative explanation or conclusion to the null hypothesis. In "*statistical testing*", a significance level (also known as an "*alpha level*") is used to determine the likelihood that the results are random. If the probability of the random results is less than the significance level, the null hypothesis may be rejected, in favor of the alternative hypothesis. The main principle of hypothesis testing is that the null hypothesis is rejected if the test statistic obtained is sufficiently unlikely under the assumption that the null hypothesis is true. In a hypothesis tests there are two types of errors. Type I error occurs when we reject a true null hypothesis, and the Type II error occurs when we fail to reject a false null hypothesis

[26] A "*t-test*" determines if there is a significant difference between the means of two independent samples or two paired (dependent) samples. It is used for quantitative variables to give an indicator of how meaningful the coefficient associated with the independent variable is in explaining variations in the dependent variable as well as the quality of the sample. It may be used to evaluate whether a single group differs from a known value (a one-sample t-test), whether two groups differ from each other (an independent two-sample t-test), or whether there is a significant difference in paired measurements based on some characteristic or variable of interest that is common to both samples (a paired, matched or dependent samples t-test). The t-test can be two-tailed, left-tailed or right-tailed. A test is one-tail if the alternative hypothesis proposes a change in the mean in only one direction (e.g. an increase). It is two-tail if the alternative hypothesis suggests a difference in the mean in either direction. The p-value is the probability of obtaining sample results as extreme or more extreme than the sample results obtained, under the assumption that the null hypothesis is true. A *t*-test for dependent samples (also called a "repeated-measures *t*-test", "paired samples *t*-test", "matched pairs *t*-test" and "matched samples *t*-test") is used to check whether the mean values of two dependent samples differ significantly from each other, under the assumption that the quantitative data is continuous, normal, independent and that there are no outliers.

	u_{1A} Use case A Result 1		u_{2A} Use case A Result 2		u_{3A} Use case A Result 3	
Cost type per unit component	u_{1AEXP} Expected Data/EUR (Group 1)	u_{1AEFF} Effective Data/EUR (Group 2)	u_{2AEXP} Expected Data/EUR (Group 1)	u_{2AEFF} Effective Data/EUR (Group 2)	u_{3AEXP} Expected Data/EUR (Group 1)	u_{3AEFF} Effective Data/EUR (Group 2)
Soil	50	60	40	35	45	46
10 Seeds	0.3 x 10	0.4 x 10 x 10	0.7 x 10	0.5 x 10	0.6 x 10	0.8 x 10
Transport	0	0	20	20	24	26
Staff for Site Preparation, Planting Site Monitoring & Maintenance	1400	1400	1500	2500	2000	1800
Equipment & Materials	250	250	200	400	250	220
TOTAL	**1703**	**1714**	**1767**	**2960**	**2325**	**2100**
Difference $D = u_{nAEFF} - u_{nAEXP}$	11		1193		-225	

Since the initial measurements (the expected and effective results) relate to the same subject matter, the sample is dependent. Therefore, paired samples T-test will be used to compare the abovementioned series of two dependent variables for the same sample set.

Let

The null hypothesis (H_0) assume that the true mean difference (μ_{nA}) is equal to zero, i.e. there is no difference :

$$H_0 ; \ \mu_{nA} = 0$$

The two-tailed alternative hypothesis (H_A) assume that μ_{nA} is not equal to zero, i.e. there is a difference :

$$H_A ; \ \mu_{nA} \neq 0$$

Such that

The number of cases is	n = 3.
The degrees of freedom is	$d_f = n - 1 = 2$.
The significance level is	α=0.05.

Here,

The sample difference mean is $\overline{\mu_{nA}}$

$$= \frac{(\mu_{1AEFF}-\mu_{1AEXP})+(\mu_{2AEFF}-\mu_{2AEXP})+(\mu_{3AEFF}-\mu_{3AEXP})}{3}$$
$$= \frac{\mu_{1AD}+\mu_{2AD}+\mu_{3AD}}{3}$$
$$= \frac{11+1193-225}{3}$$
$$= \frac{979}{3}$$

The sample standard deviation is S_D

$$= \sqrt{\frac{\left(11-\frac{979}{3}\right)^2+\left(1193-\frac{979}{3}\right)^2+\left(-225-\frac{979}{3}\right)^2}{2}}$$
$$= \sqrt{\frac{\left(-\frac{946}{3}\right)^2+\left(\frac{2600}{3}\right)^2+\left(-\frac{1654}{3}\right)^2}{2}}$$

Let

The test statistic be t

$$= \frac{\overline{\mu_{nA}}-0}{S_D/\sqrt{n}}$$
$$= \frac{\frac{979}{3}-0}{S_D/\sqrt{3}}$$

And

$$P=2 \cdot P_r(T > |t|)$$

Such that

D	: Difference
n	: The sample size
$\overline{\mu_{nA}}$	: The sample mean of the differences
S_D	: The sample standard deviation of the differences
T	: The critical value of a t-distribution with (n − 1) degrees of freedom
t	: The t-statistic (t-test statistic) for a paired sample t-test
p	: The p-value (probability value) for the t-statistic.

The same test may be used vis-à-vis a group of projects within the same category, or all the projects combined.

(ii) Regression Analysis

A regression analysis is a statistical analysis technique used to analyze the relationship between one or more independent variables and a dependent variable. It can be used to model and analyse the relationship between two or more variables to assess how correlated or associated the two variables are (e.g. between the number of units for a component and the cost differences). It helps identify the strength and direction of the relationship between variables, make predictions, and estimate the impact of changes in one variable on another. Regression analysis is a tool for predicting outcomes and making decisions based on data. In simple linear regression, there is only one independent variable and one dependent variable.

For example, to

Average cost of seeds (Y)	Number of units deployed (X)
0.3	10
0.25	100
0.22	500

Let the regression equation be : $Y = a + bX + e$

Such that

X : Independent variable (X)
Y : Dependent variable
a : y-intercept
b : slope of the regression line
e : error term

Here,

Sum of X (Σ_X) = 0.77

Sum of Y (Σ_Y) = 610

Mean X ($\overline{X}$) $= \frac{\sum_{i=1}^{n} X_i}{n}$
= 0.257

Mean Y ($\overline{Y}$) $= \frac{\sum_{i=1}^{n} Y_i}{n}$
= 203.333

Sum of products (SP) $= \sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$
= -18.5667

Sum of squares (SSX) $= \sum_{i=1}^{n}(X_i - \overline{X})^2$
= 0.0033

slope of the regression line (b) $= \frac{SP}{SSX}$
$= \frac{[n(\Sigma XY) - (\Sigma X)(\Sigma XY)]}{[n(\Sigma X^2) - (\sum X)^2]}$
= -18.57/0
= -5683.673

Standard error (e) = 0.027

y-intercept (a) $= \overline{Y} - b\overline{X}$
$= \frac{[(\Sigma_Y)(\Sigma X^2) - (\Sigma X)(\Sigma XY)]}{[n(\Sigma X^2) - (\sum X)^2]}$
= 203.33 - (-5683.67x0.26)
= 1662.143

Y = -5683.673X + 1662.143

F -value = 3.455

Probability F-value = 0.314

Correlation coefficient (r) = -0.881

$0.7 < |r| \leqq 1$ so there is a moderate positive correlation between X and Y.

r square = 0.776

77.555% of the variation in Y is explained by X.

(iii)Chi-square Test : Association and Homogeneity

A chi-square test is a non-parametric statistical test that does not assume any particular distribution of the data. It can be used for different purposes, such as testing for homogeneity across multiple categorical variables or to determine whether there is a significant association. It may also be used to compare the expected and observed frequencies of data in each category, calculated based on a specified hypothesis of independence between the variables. The test essentially calculates a chi-square statistic, then compared to a critical value from a chi-square distribution based on the degrees of freedom (DF) and level of significance (α). The test assumes certain conditions, such as the absence of outliers, independent observations, and expected cell frequencies greater than or equal to five. Violation of these assumptions may affect the validity of the results. Finally, the test only determines whether there is an association between the variables, but does not establish causation. For example, it would be possible to test if there is an association between the different countries and the higher global category result (R_n).

Chi-square test : Higher global category result (R_n) vs Country				
	Country A	Country B	Country C	**Total**
Lower global category result (R_n)	12	20	15	47
Expected	0.47 x 0.35 x 100 = 16.45 if independent variables	0.47 x 0.40 x 100 = 18.8 if independent variables	0.47 x 0.25 x 100 = 11.75 if independent variables	
Higher global category result (R_n)	23	20	10	53
Expected	0.53 x 0.25 x 100 = 18.55 if independent variables	0.53 x 0.25 x 100 = 21.2 if independent variables	0.53 x 0.50 x 100 = 13.25 if independent variables	
Total	35	40	25	100

Let

The null hypothesis be expressed as:

H_0 = The two categorical variables are independent variables.

The alternative hypothesis be expressed as:

H_A = There is an association between the two categorical variables.

The level of significance be :

$$\alpha = 0.05$$

If the calculated chi-square value is greater than the critical p value, there is a significant association between the variables.

A contrario, if it is less than the critical value, there is no significant association between the variables (confirmation of independence of the variables).

Such that

$$x^2 = \frac{(12-16.45)^2}{16.45} + \frac{(20-18.8)^2}{18.8} + \frac{(15-11.75)^2}{11.75} + \frac{(23-18.55)^2}{18.55} + \frac{(20-21.2)^2}{21.2} + \frac{(10-13.25)^2}{13.25}$$
$$= 4.11$$

DF = (number of rows - 1) x (number of columns - 1)
= (2-1) x (3-1) = 2

P-value = 0.128

P < α so H_0 is rejected, i.e. there is a significant association between the variables.

(iv) One-way ANOVA N-Sample Independent Test : Utility Cases & Efficiency

The one-way Analysis of Variance test ("ANOVA test") for a single categorical independent variable is a parametric test used to examine if there is a statistically significant difference among the means of three or more populations.

It is based on the principle that the total variation in the data can be decomposed into: the variability between the sample means themselves (the "treatment" variation), and the variability of the data within each sample (the "error" variation). The ANOVA test assumes that the data meets certain assumptions, such as the normality, the homogeneity of variance, the randomness of the data sample, and the independence of observations.

For example, if the efficiency of each global utility case (U_A, U_B, U_C) were to be compared by taking a random sample :

	U_A = tackling deforestation	**U_B = renewable energy**	**U_C = waste management**
p_1	0.71	0.79	0.66
p_2	0.60	0.68	0.62
p_3	0.61	0.72	0.61

Hypotheses :

H_0 : $\mu_1=\mu_2=\mu_3=\mu_k$ (the means are all equal.)

H_A : the means are not all equal.

where k = the number of independent comparison groups.

If p-value ≤ α, reject H_0 in favour of H_A.

The results of the sample data are significant. There is sufficient evidence to conclude that the null hypothesis H0 is an incorrect belief, and that the alternative hypothesis Ha is most likely correct.

If p-value>α, do not reject H_0.

The results of the sample data are not significant. There is not sufficient evidence to conclude that the alternative hypothesis H_A may be correct.

	$\mathbf{U_A}$	$\mathbf{U_B}$	$\mathbf{U_C}$
	Group 1	Group 2	Group 3
Sample Size (n_n)	$n_1 = 3$	$n_2 = 3$	$n_3 = 3$
Group average ($\bar{x}_n$)	$\bar{x}_1 = 1.92/3$ =0.64	$\bar{x}_2 = 2.19/3$ =0.73	$\bar{x}_3 = 1.89/3$ =0.63

Let

K = number of groups
= 3

N = overall sample side, including all the groups
$= n_1 + n_2 + n_3$
= 3+3+3
= 9

$\bar{\bar{x}}$ = overall average
$= [(3x0.64)+ (3x0.73)+ (3x0.63)]/9$
= 0.667

SST $= n_1 \text{ x } (\bar{x}_1 - \bar{\bar{x}})^2 + n_2 \text{ x } (\bar{x}_2 - \bar{\bar{x}})^2 + n_3 \text{ x } (\bar{x}_3 - \bar{\bar{x}})^2$
$= 3 \text{ x } (0.64 - 0.667)^2 + 3 \text{ x } (0.73 - 0.667)^2 + 3 \text{ x } (0.63 - 0.667)^2$

$$MST = \frac{SST}{(k-1)}$$
$$= \frac{3 \text{ x } (0.64-0.667)^2 + 3 \text{ x } (0.73-0.667)^2 + 3 \text{ x } (0.63-0.667)^2}{(3-1)}$$

$$SSE = (n_1 - 1) \text{ x } {s_1}^2 + (n_2 - 1) \text{ x } {s_2}^2 + (n_3 - 1) \text{ x } {s_3}^2$$
$$= (3\text{-}1) \text{ x } 0.64^2 + (3\text{-}1) \text{ x } 0.73^2 + (3-1) \text{ x } 0.63^2$$

$$MSE = \frac{SSE}{(n-k)}$$
$$= \frac{(3-1) \text{ x } 0.64^2 + (3-1) \text{ x } 0.73^2 + (3-1) \text{ x } 0.63^2}{(9-3)}$$

The test statistic for testing H_0 : $\mu_1=\mu_2=\mu_3=\mu_k$ is :

$$F = \frac{\text{treatment variation}}{\text{error variation}}$$
$$= \frac{MST}{MSE}$$
$$= \frac{\frac{3 \text{ x } (0.64-0.667)^2 + 3 \text{ x } (0.73-0.667)^2 + 3 \text{ x } (0.63-0.667)^2}{(3-1)}}{\frac{(3-1) \text{ x } 0.64^2 + (3-1) \text{ x } 0.73^2 + (3-1) \text{ x } 0.63^2}{(9-3)}}$$

With df_1 = k -1 (degrees of freedom for the F-distribution)
= 3-1
=2

And df_2 = n−k (k : sample size per group ; n : total sample size for all groups)
= 9-3
=6

Source	DF	Sum of Square	Mean Square	F Statistic	P-value
Treatment	2	0.0182	0.0091	3.64	0.09223
Error	6	0.015	0.0025		
Total	8	0.0332	0.00415		

The test statistic F = 3.640004, which is in the 95% region of acceptance: [-∞ : 5.1433]

p-value>α, so H_0 is accepted.

The difference between the averages of all groups is not big enough to be statistically significant, i.e. the averages of all groups assumed to be equal.

(v) Newcomb Benford's Law : Transparency and Prevention of Accounting Data Fraud

In practice, corruption remains a barrier to sustainable development goals and pro-environmental projects.[27] Benford's Law may be applied to analyse datasets related to the metrics in the Model. Since the core data used in the Model relies partly on existing accounting and financial data, it is suggested that for the purposes of detecting fraud relating to the individual input variable (x_n), i.e. the declaration by companies of the costs engaged for their '*green project'*, the Newcomb Benford's Law be used following the formula :

$$\text{Prob}\ (D_1 = d) = log_{10}\left(1 + \frac{1}{d}\right) \text{ for all d = 1, 2, . . ., 9}$$

This method is already highly used for the cited purpose in diverse contexts, including fraud detection in relation to quantitative data (e.g. accounting data, scientific data, amongst others). [28] It would also be in lines with the anticorruption, anti-money laundering and anti-bribery rules, transparency requirements, and accounting standards.[29]

---

[27] Henderson, W.T., "*Building an effective and cost-efficient anti-corruption program", 2011,* Corporate Finance Review, 16(1), p.11.; Aldaz, M., et al. "*Non-financial reports, anti-corruption performance and corporate reputation*", 2015, Revista brasileira de gestão de negócios, 17, pp.1321-1340.; Xu, G. et al; "*How does anti-corruption affect corporate innovation? Evidence from recent anti-corruption efforts in China*", 2017, Journal of comparative Economics, 45(3), pp.498-519.; Sullivan, J.D., "*The moral compass of companies: Business ethics and corporate governance as anti-corruption tools*", 2009. Focus, 7, p.47791.; Álvarez Etxeberria, I. et al. "*The social reputation of European companies: Does anti-corruption disclosure affect stakeholders' perceptions?*", 2018, Corporate Social Responsibility and Environmental Management, 25(5), pp.713-721.; Wei, Y. et al. "*Can anti-corruption improve the quality of environmental information disclosure?*" 2022, Environmental Science and Pollution Research, pp.1-15.; Owusu, E.K., et al. "*Impacts of anti-corruption barriers on the efficacy of anti-corruption measures in infrastructure projects: Implications for sustainable development*", 2020, Journal of cleaner production, 246, p.119078.; Williams, A. et al. "*Deciding over nature: Corruption and environmental impact assessments*", 2017, Environmental Impact Assessment Review, 65, pp.118-124.; Nest, M. et al. "*Lobbying, corruption and climate finance: The stakes for international development*", 2021,. U4 Issue.

[28] Ascioglu, A., et al. L., "*Analysis of Sustainability Reports for Top 20 Companies in the S&P 500 Index*", 2022, *The Journal of Impact and ESG Investing*, *2*(3), pp.82-94.; Cole, M.A., et al. L., "*Testing the emission reduction claims of CDM projects using the Benford's Law*", 2020, *Climatic Change*, *160*, pp.407-426.; Diekmann, A., "*Not the first digit! using benford's law to detect fraudulent scientific data*", 2007, *Journal of Applied Statistics*, *34*(3), pp.321-329.; Coracioni, A.T., "*Testing of published information on greenhouse gas emissions. Conformity analysis with the Benford's Law Method*", 2020, Audit Financiar, 18(4), pp.821-830.; Stoerk, T., "*Statistical corruption in Beijing's air quality data has likely ended in 2012*", 2016, Atmospheric Environment, 127, pp.365-371.; Durtschi, C., et al. "*The effective use of Benford's law to assist in detecting fraud in accounting data*", 2004, Journal of forensic accounting, 5(1), pp.17-34.; Burgos, A. et al. "*The Newcomb–Benford law: Scale invariance and a simple Markov process based on it*", 2021, American Journal of Physics, 89(9), pp.851-861.; Cerioli, A., et al. "*Newcomb–Benford law and the detection of frauds in international trade*", 2019, *Proceedings of the National Academy of Sciences*, *116*(1), pp.106-115.

[29] 2021/0250(COD) ; 2021/0240(COD) ; 2021/0239(COD); Directive 2015/849/EU; Directive 2018/843/EU; Regulation (EU) 2015/847; Article L123-12 of the French Commercial Code; French law n°2016-1691 ("*loi Sapin II*"); French Anti-Corruption Agency (*Agence Française Anticorruption - AFA*), "*Corporate anti-corruption accounting controls : A Practical Guide*", April 2022; US Foreign Corrupt Practices Act 1977, as amended, 15 U.S.C. §§ 78dd-1, et seq. (FCPA); UK Bribery Act 2010 ; South Korea's Act on Preventing Bribery of Foreign Public Officials in International Business Transactions 1999 (FBPA); OECD, "*Convention on Combating Bribery of Foreign Public Officials in International Business Transactions*", OECD/LEGAL/0293; UK's Money Laundering and Terrorist Financing (Amendment) (No. 2) Regulations 2022 n°860; US Anti-Money Laundering Act 2020 (AMLA); US Corporate Transparency Act 2021 (CTA); Pub. L. No. 116-283 (January 1, 2021) §6403;

Further, using Benford's Law can help identify anomalies and ensure data reliability, which is crucial for effective environmental policy and decision-making. Understanding which data sets conform to expected patterns can help prioritise areas for intervention and resource allocation.

CONCLUSION

In conclusion, environmental awareness continues to play a pivotal role in shaping consumer choices. Despite robust legal and regulatory frameworks, the need for effective enforcement persists, hindered by the lack of a universal standard. This paper emphasizes corporate self-regulation and transparent reporting as crucial elements in addressing these challenges. By proposing an innovative mathematical model, it bridges interdisciplinary approaches to enhance practical applications in environmental impact assessment. The model's integration of quantitative and qualitative variables, alongside a cost-analysis approach, offers a valuable tool for evaluating environmental efficiency and identifying potential fraud. Additionally, it facilitates cost-risk assessments and supports data-driven decision-making for a wide range of stakeholders. Ultimately, this approach underscores the principles of transparency, precaution, and international solidarity, reinforcing the commitment to sustainable practices and the circular economy. The benefits of the suggested tool and approach remains its repeatability throughout different sectors and industries, as well as its accessibility. Flexibility is particularly appreciable in the vast context of sustainability whereby different types and scopes of projects contribute to the dynamic and complex ecosystem.[30] Despite the usefulness of the tool, some of the major obstacles to this mechanics remain confidentiality, business secrecy, access and exploitation of sensitive data in a regulated field.

31 USC §5336.

[30] Majda, A.J., "*Challenges in climate science and contemporary applied mathematics. Communications on Pure and Applied Mathematics*", 2012, 65(7), pp.920-948. ("*predicting and understanding the seasonal, yearly, decadal and centennial impacts of climate change for issues ranging from extreme weather events to sea level rise to the evolving extent of deserts involves assessing the impacts of climate change on a variety of significant temporal and spatial scales*")

BIBLIOGRAPHY

Aldaz, M., et al., 'Non-Financial Reports, Anti-Corruption Performance and Corporate Reputation' (2015) 17 Revista brasileira de gestão de negócios 1321-1340

Álvarez Etxeberria, I., et al., 'The Social Reputation of European Companies: Does Anti-Corruption Disclosure Affect Stakeholders' Perceptions?' (2018) 25(5) Corporate Social Responsibility and Environmental Management 713-721

Article 3(3) of the Treaty on European Union (TEU)

Articles 11 and Article 191 of the Treaty on the Functioning of the European Union (TFEU)

Article 37 of the European Charter of Fundamental Rights

Ascioglu, A., et al., 'Analysis of Sustainability Reports for Top 20 Companies in the S&P 500 Index' (2022) 2(3) The Journal of Impact and ESG Investing 82-94

Balvanera, P., et al., 'Essential Ecosystem Service Variables for Monitoring Progress towards Sustainability' (2022) 54 Curr. Opin. Environ. Sustain. 101152

Brown, M.T., et al., 'Predicting National Sustainability: The Convergence of Energetic, Economic and Environmental Realities' (2009) 220(23) Ecological Modelling, ELSEVIER 3424-3438

Bulman v Fenwick & Co [1894] 1 QB 179

Climate Benchmarks Regulation EU 2019/2089 and 2020/1818

Cole, M.A., et al., 'Testing the Emission Reduction Claims of CDM Projects Using the Benford's Law' (2020) 160 Climatic Change 407-426

Copernicus Climate Change Service, ECMWF, European Commission, 'Essential Climate Variables for Assessment of Climate Variability from 1979 to Present' (7 November 2018, updated 5 October 2023) https://cds.climate.copernicus.eu/cdsapp#!/dataset/ecv-for-climate-change?tab=overview

Council of the European Union, 'Proposal for a Directive of the European Parliament and of the Council Amending Directives 2005/29/EC and 2011/83/EU as Regards Empowering Consumers for the Green Transition through Better Protection against Unfair Practices and Better Information' 2022/0092(COD)

CPC Group Ltd v Qatari Diar Real Estate Investment Co [2010] EWHC 1535 (Ch)

Das, A., et al., ‘How do Companies Measure and Forecast Environmental Impacts When Experimenting with Circular Business Models?’ (2022) 29 Sustainable Production and Consumption 273-285

Directive (EU) 2022/2464

Directive 2015/849/EU

Directive 2018/843/EU

Directive on Units of Measurement 80/181/EEC

Diekmann, A., ‘Not the First Digit! Using Benford's Law to Detect Fraudulent Scientific Data’ (2007) 34(3) Journal of Applied Statistics 321-329

Dündar, S., et al., ‘Mathematical Modelling at a Glance: A Theoretical Study’ (2012) 46 Procedia - Social and Behavioral Sciences

Durtschi, C., et al., ‘The Effective Use of Benford’s Law to Assist in Detecting Fraud in Accounting Data’ (2004) 5(1) Journal of Forensic Accounting 17-34

Electricity Generation Corporation (t/as Verve Energy) v Woodside Energy Ltd (2014) 251 CLR 540

European Commission, ‘Corporate Sustainability Reporting’ https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en

European Council, Council of the European Union, ‘Timeline - European Green Deal and Fit for 55’ https://www.consilium.europa.eu/en/policies/green-deal/timeline-european-green-deal-and-fit-for-55/

European Parliamentary Research Service (EPRS), G. RAGONNAUD, Briefing, ‘EU Legislation in Progress, “Green Claims” Directive Protecting Consumers from Greenwashing’ (10 May 2023) https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2023)753958

European Parliamentary Research Service (EPRS), N. SAJN, ‘Empowering Consumers for the Green Transition’ (16 June 2022) https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2022)733543

European Parliament, Policy Department for Citizens' Rights and Constitutional Affairs Directorate-General for Internal Policies, Prof. Dr. Lorenzo SQUINTANI, 'Study on the Harmonisation of EU Environmental Law' PE 705.033 (February 2022)

European Parliament, Press Release, 'EU to Ban Greenwashing and Improve Consumer Information on Product Durability' (19 September 2023) https://www.europarl.europa.eu/news/en/press-room/20230918IPR05412/eu-to-ban-greenwashing-and-improve-consumer-information-on-product-durability

Figge, F., Hahn, T., 'The Sustainable Value Approach: A Model to Assess the Monetary Contribution of Companies to Sustainability' (2004) 1(1) Ecological Economics

Fowler, A.C., 'Mathematical Models in the Applied Sciences' (1997) Cambridge University Press

French Anti-Corruption Agency (Agence Française Anticorruption - AFA), 'Corporate Anti-Corruption Accounting Controls: A Practical Guide' (April 2022)

French Consumer Code (Articles L121-1 to L121-4, L132-1 to L132-2, L531-6, R531-3 and L521-7)

French law n°2008-3

French law n°2016-1691 ('loi Sapin II')

Glenn, M., et al., 'An Introduction to Mathematical Modelling' (2008)

Grimm, J.H., Hofstetter, J.S., et al., 'Exploring Sub-Suppliers' Compliance with Corporate Sustainability Standards' (2016) 112 Journal of Cleaner Production 1971-1984

Guest, R., 'The Economics of Sustainability in the Context of Climate Change: An Overview' (2010) 45(4) Journal of World Business 1090-9516 https://doi.org/10.1016/j.jwb.2009.08.008
Henderson, W.T., 'Building an Effective and Cost-Efficient Anti-Corruption Program' (2011) 16(1) Corporate Finance Review 11

Ibáñez-Forés, V., et al., 'Sustainability Reports as a Tool for Measuring and Monitoring the Transition towards the Circular Economy of Organisations: Proposal of Indicators and Metrics' (2022) 320 Journal of Environmental Management 115784 https://doi.org/10.1016/j.jenvman.2022.115784

International Organization of Securities Commissions, 'IOSCO Endorses the ISSB's

Sustainability-Related Financial Disclosures Standards' IOSCO/MR/19/2023 (25 July 2023)
International Trade Administration, 'EU Country Commercial Guide' (11 August 2022) https://www.trade.gov/country-commercial-guides/eu-labelingmarking-requirements

ISO 14007

ISO 14008

ISO 14097

ISO 32210

Jankalová, M., et al., 'Sustainability Assessment Using Economic Value Added' (2020) 12(1) Sustainability 318 https://doi.org/10.3390/su12010318

Joint CEOS/CGMS Working Group Climate, Schulz, J., et al., 'Statement of Space Agency Contributions in Support of the Paris Agreement' (16 November 2018)

Khan, S.A.R., et al., 'Measuring the Impact of Renewable Energy, Public Health Expenditure, Logistics, and Environmental Performance on Sustainable Economic Growth' (2020) 28(4) Sustainable Development 833-843

Lenton, Timothy M., et al., 'Quantifying the Human Cost of Global Warming' (2023) 1-11 Nature Sustainability

Majda, A.J., 'Challenges in Climate Science and Contemporary Applied Mathematics' (2012) 65(7) Communications on Pure and Applied Mathematics 920-948

Measuring Instruments Directive 2014/32/EU

Moore, J.E., et al. "Developing a comprehensive definition of sustainability", 2017, Implementation Science, 12(1), pp.1-8.

Munda, G., 'Measuring Sustainability: A Multi-Criterion Framework' (2005) 7 Environment Development and Sustainability 117-134

Mukherji, A., et al., 'Synthesis Report of the IPCC Sixth Assessment Report (AR6)' (2023) United Nations

MUR Shipping BV v RTI Ltd [2022] EWCA Civ 1406, [2022] WLR(D) 427, [2023] 1 All ER (Comm) 501, [2023] Bus LR 355, 205 Con LR 1

Munda, G., ‘Measuring Sustainability: A Multi-Criterion Framework. Environment Development and Sustainability’ (2005) 7, 117-134

Nest, M. et al. “Lobbying, corruption and climate finance: The stakes for international development”, 2021,. U4 Issue.

Non-Financial Reporting Directive (NFRD) 2014/95/EU

Owusu, E.K., et al. “Impacts of anti-corruption barriers on the efficacy of anti-corruption measures in infrastructure projects: Implications for sustainable development”, 2020, Journal of cleaner production, 246, p.119078.;

Plant, A.L. et al. “Reproducibility and replicability in science, a metrology perspective”, 15.06.2018, National Academies of Sciences, Engineering and Medicine Committee on Reproducibility and Replicability in Science Report.

Regulation 2018/858/EU

Regulation 2019/2144/EU

Regulation 2018/858/EU

Regulation 78/2009/EC

Regulation 79/2009/EC

Regulation 661/2009/EC

Regulation 631/2009/EC

Regulation 406/2010/EU

Regulation 672/2010/EU

Regulation 1003/2010/EU

Regulation 1005/2010/EU

Regulation 1008/2010/EU

Regulation 1009/2010/EU

Regulation 19/2011/EU

Regulation 109/2011/EU

Regulation 458/2011/EU

Regulation 65/2012/EU

Regulation 130/2012/EU

Regulation 347/2012/EU

Regulation 351/2012/EU

Regulation 1230/2012/EU

Regulation 2015/166/EU

Sustainable Finance Disclosure Regulation EU 2019/2088

Sullivan, J.D., "The moral compass of companies: Business ethics and corporate governance as anti-corruption tools", 2009. Focus, 7, p.47791.;

Taxonomy Regulation EU 2020/852

The Economist, "Why tackling global warming is a challenge without precedent", 23.04.2020, https://www.economist.com/schools-brief/2020/04/23/why-tackling-global-warming-is-a-challenge-without-precedent

The State of the Netherlands (Ministry of Economic Affairs and Climate Policy) v Stichting Urgenda, The Supreme Court of the Netherlands, 19/00135

Tucker, D.T., Stanford University, "What is Sustainability? A definition by environmental scientist Pamela Matson", 05.21.2018, https://earth.stanford.edu/news/what-sustainability-definition-environmental-scientist-pamela-matson
UBH (Mechanical Services) Ltd v. Standard Life Assurance Co. T.L.R., 13 Nov. 1986 (Q.B.)

UCL (Department of Geography) and Center for Global Environmental Research, National Institute for Environmental Studies CARTER, T.R., et al. "IPCC Technical guidelines for assessing climate change impacts and adaptations." 1994, Intergovernmental Panel on Climate Change, Part of the IPCC Special Report to the First Session of the Conference of the Parties to the UN Framework Convention on Climate Change, World Meteorological

Organiwation/United Nations Environment Programme, CGER-I015-94

United States Environmental Protection Agency, “Learn About Sustainability”, https://www.epa.gov/sustainability/learn-about-sustainability#what

United Nations Sustainable Development Goals (SDG)

United Nations Climate Change, “Overview; Implementation of systematic observation”, https://unfccc.int/es/node/15979

Vereniging Milieudefensie v Royal Dutch Shell PLC, the Hague District Court, C/09/571932 / HA ZA 19-379, ECLI:NL:RBDHA:2021:5337

Vigneau, L., et al. “How do firms comply with international sustainability standards? Processes and consequences of adopting the global reporting initiative”, 2015, Journal of business ethics, 131, pp.469-486.

Watling, A. et al. “Attitudes towards Sustainability: A Quantitative Study of Sustainable Ålidhem”. 2011, Umea Universitet, Umea, Sweden. http://umu.diva-portal.org/smash/record.jsf?pid=diva2%3A430152&dswid=-6754

Williams, A. et al. “Deciding over nature: Corruption and environmental impact assessments”, 2017, Environmental Impact Assessment Review, 65, pp.118-124.

World Health Organization (WHO), “WHO issues urgent call for global climate action to create resilient and sustainable health systems” 24.05.2023, https://www.who.int/news/item/24-05-2023-wha76-strategic-roundtable-on-health-and-climate

Xu, G. et al; “How does anti-corruption affect corporate innovation? Evidence from recent anti-corruption efforts in China”, 2017, Journal of comparative Economics, 45(3), pp.498-519.

Yewbelle v London Green [2007] EWCA Civ 475